DIVISION AVIONIQUE

Présentation générale

Présenté par Nathan Armishaw

REV A
Septembre 2025

UNIVERSITÉ LAVAL

Table des matières

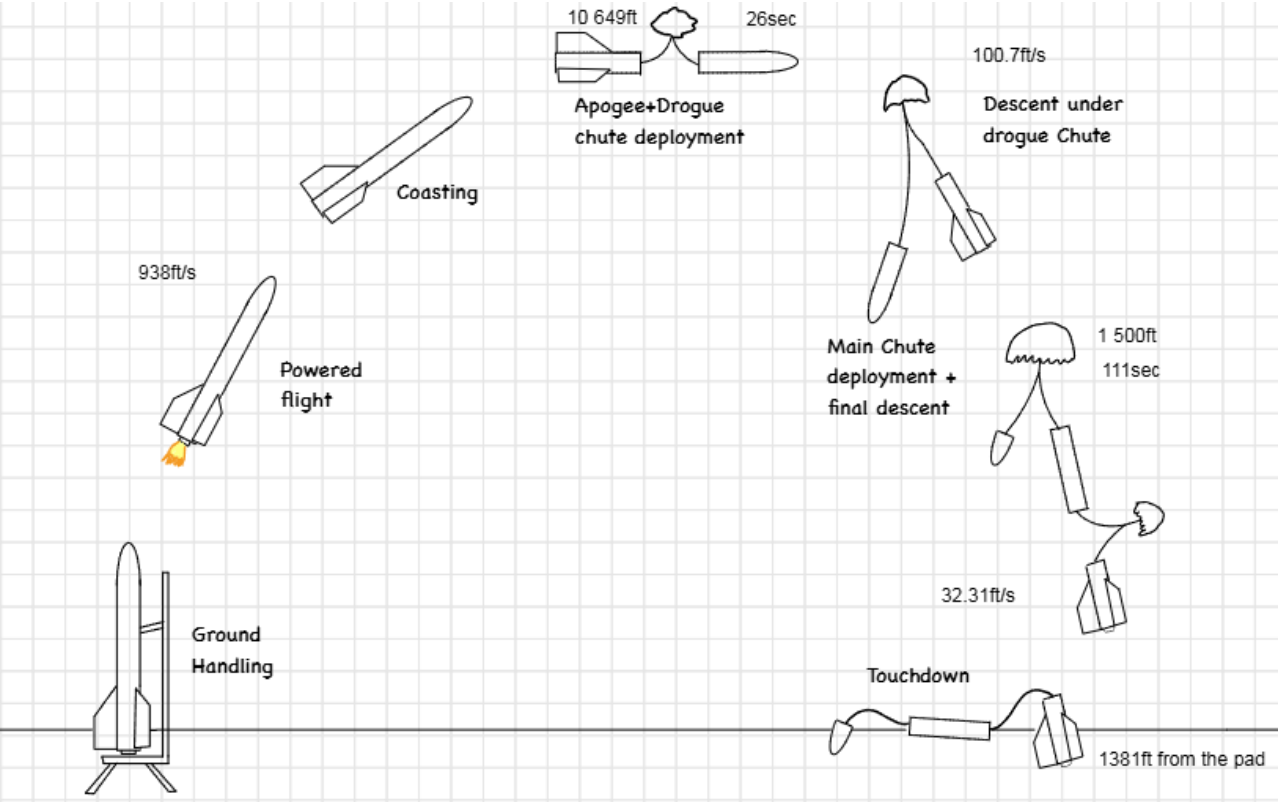
- CONOPS (Concept of Operations)
- Requis généraux
- Compétitions
- Appareils COTS
- Nos appareils SRAD
- Système LC 2026

UNIVERSITÉ LAVAL

CONOPS

(MISSION CONCEPT OF OPERATIONS)

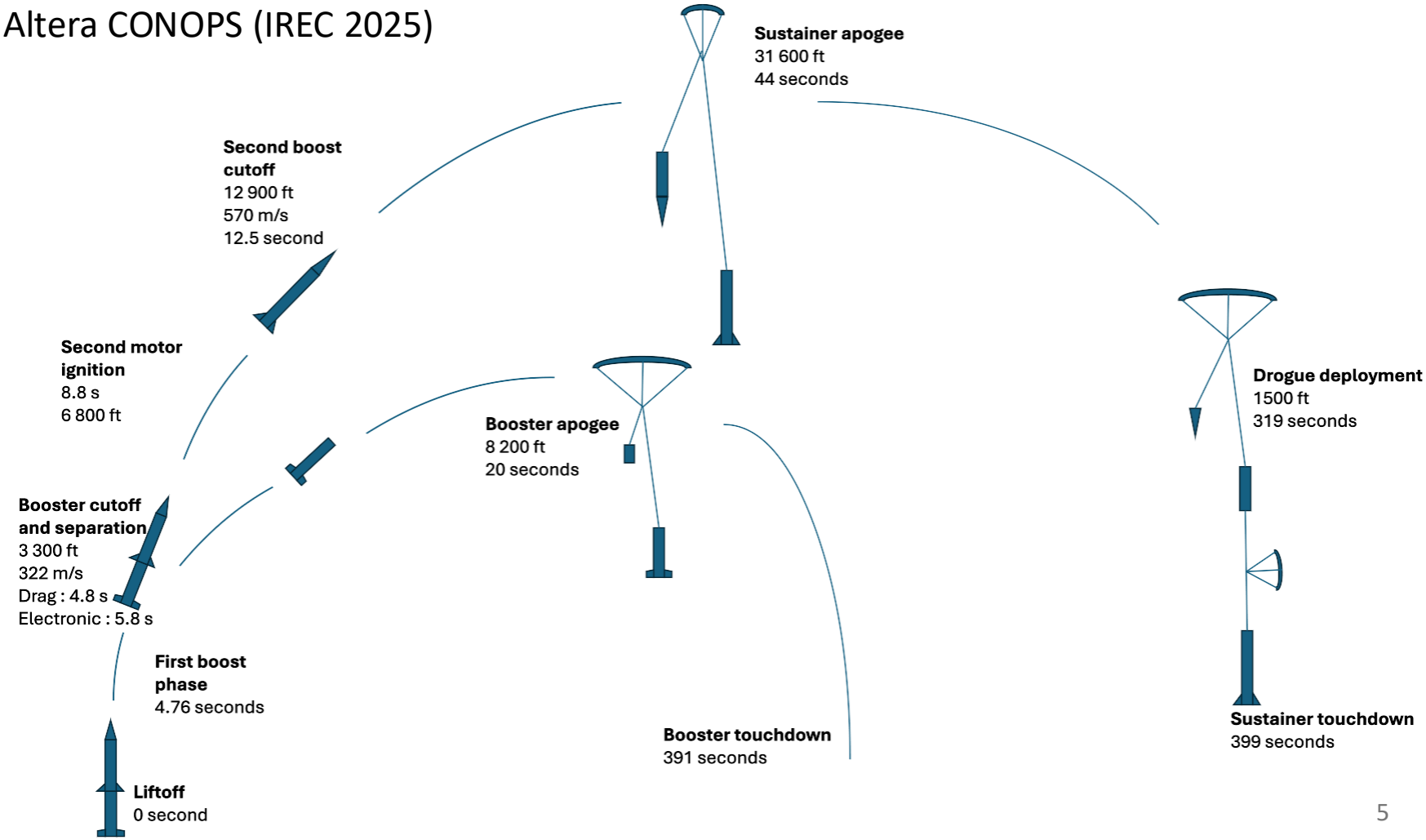
Mérope CONOPS (Spaceport America Cup 2024)



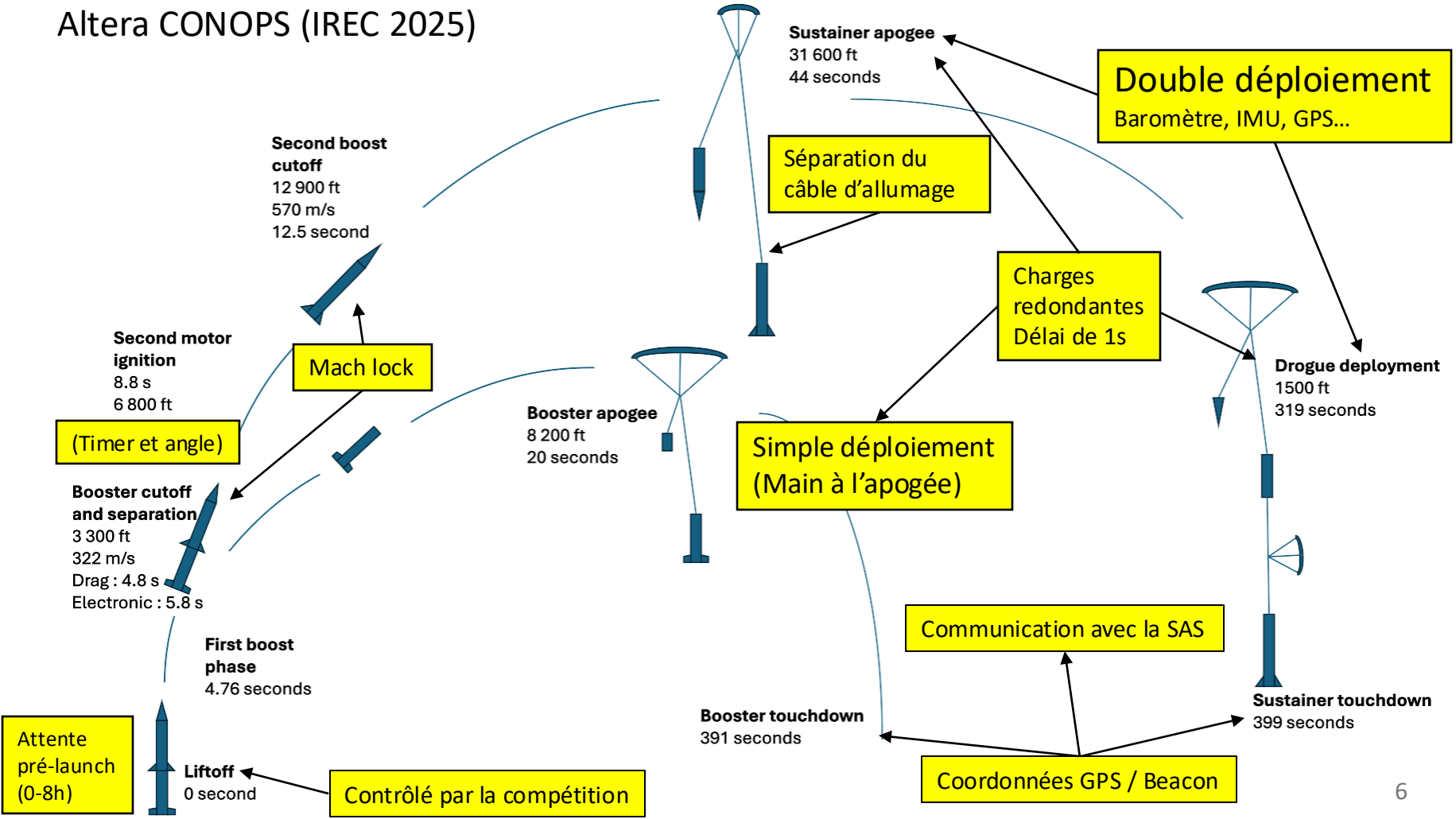
CONOPS

- Résume les étapes critiques de la mission
- Haut niveau (peu de détails)
- Point de vue du vol général
- On en dérive les requis avioniques

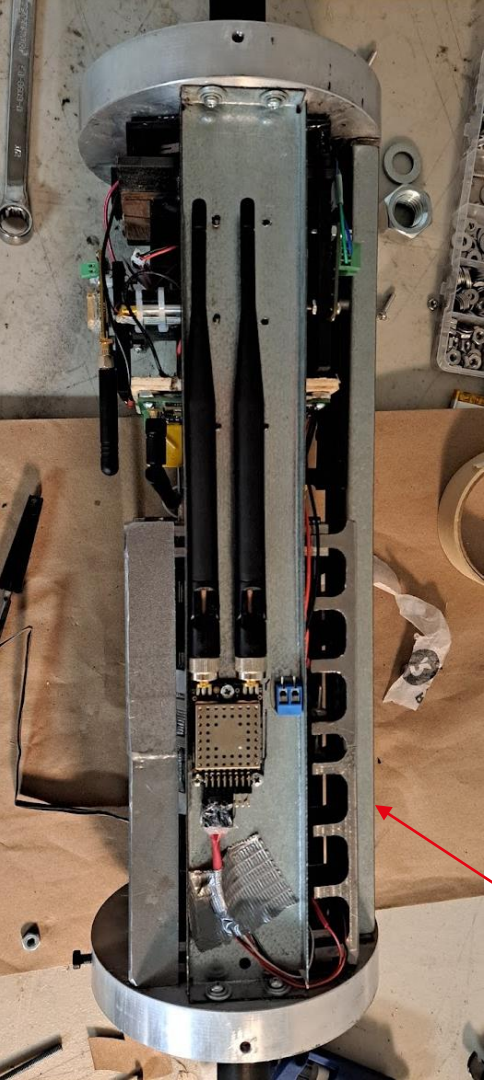
Altera CONOPS (IREC 2025)



Altera CONOPS (IREC 2025)



REQUIS GÉNÉRAUX



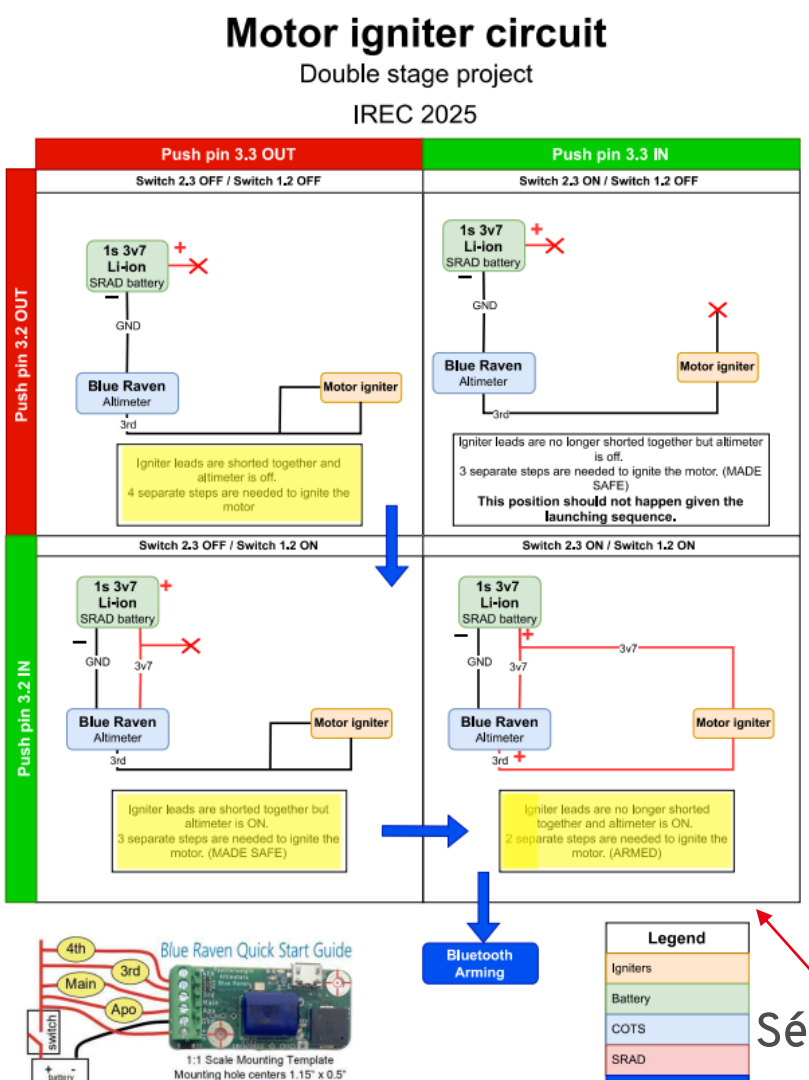
Requis généraux

Comment est-ce qu'on traduit ça en baie avionique?

- Sécurité
- Redondance
- Communication
- Autonomie de batterie
- Vérifications
- Dimensions de la baie avionique
- Robustesse

Exemple de baie « fonctionnelle »
Spoiler alert; on peut faire beaucoup mieux

UNIVERSITÉ LAVAL



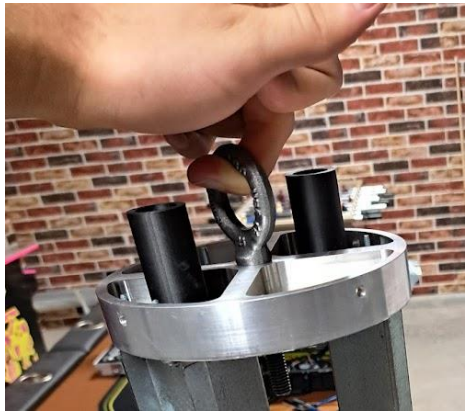
Sécurité

Pour que rien n'explose à 5 pouces de notre visage…

- Composantes à haute densité d'énergie (batteries, poudre noire, moteurs)
- Made Safe (2 étapes avant l'ignition)
- Armed (1 étape avant l'ignition)
- Armement sécuritaire (à distance)
- Fail safe (Si rien ne fonctionne, est-ce que ce sera sécuritaire?)

Séquence d'armement du sustainer d'Altera

Redondance



Pas juste « Au cas où »

2 charges par séparation

2 appareils différents par séparation

1 batterie séparée par appareil

Pas de redondance pour l'allumage des moteurs!!!

COMPÉTITIONS







UNIVERSITÉ LAVAL

APPAREILS COTS

(COMMERCIAL OFF THE SHELF)

CE QU'ON A EN STOCK SEULEMENT

Famille Featherweight Altimeters

	Blue Raven	1" Diameter Blue Raven	Blue Jay2 Plus	Blue Jay2
Price	$175	$195	$100	$80
User interface				
Custom app for iPhone and Android				
Live output continuity and battery voltages	Yes	Yes	Yes	Yes
Live sensor health	Yes	Yes	Yes	Yes
Software arming option	Yes	Yes	Yes	Yes
Simplified settings for common applications	Yes	Yes	Yes	Apogee and later
Full custom deployment logic	Yes	Yes	Yes	Apogee and later
Extensive post-flight summary data	Yes	Yes	Yes	Yes
Graph recorded flight data	Yes	Yes	low-rate	low-rate
Export recorded flight data	Yes	Yes	low-rate	low-rate
Remote ground test with slo-mo video	Yes	Yes	Yes	Yes
Beeper	Yes	Yes	Yes	Yes
LEDs	3	3	2	2
Outputs				
Single screw terminal per output (return side)	4			
Mounting hole terminal		2		
Solder pad per output (return side)		2		
Dual terminal per ouptut (+ and return)			2	2
Tolerates output short or disconnected battery for 6 seconds	Yes	Yes	No	No
Current-limited outputs	Yes	Yes	Yes	Yes
Hardware support for possible future servo output control	3 channels	No	No	No
Installation				
Built-in magnetically activated power switch	No	Yes	Yes	Yes
Install in any orientation	Yes	Yes	Yes	Yes
Footprint	1.8" x 0.8"	1.0" dia	1.45" x 0.65"	1.45" x 0.65"
Smallest diameter rocket	24mm	29mm	18mm	18mm
Length in Av-bay	1.8"	0.45"	1.45"	1.45"
Power consumption	20 mA	20 mA	15 mA	15 mA
Input power compatibility	3V-12V	1s LiPo cell	3V-12V	3V-12V
Sensor and data recording				
3-Axis high-accuracy accelerometer, 32 G range	Yes, 500Hz +/- 32Gs	Yes, 500Hz +/- 32Gs	16G Axial and lateral at 50 Hz	16G Axial and lateral at 50 Hz
3-Axis High G Range Accelerometer	Yes, 500 Hz +/- 400 Gs	Yes, 500 Hz +/- 400 Gs	320G Axial and lateral at 50 Hz	320G Axial and lateral at 50 Hz
3-Axis Gyro, +/- 2100 deg/sec range	Yes	Yes	raw data not recorded	raw data not recorded
Barometric sensor maximum altitude	~100 kft	~100 kft	~100 kft	~100 kft
500 Hz Accel, gyro, quaternion, current, aux	Yes	Yes	No	No
50 Hz Altitude, 3-DOF position, voltages, temperature, tilts, flight events	Yes	Yes	Yes	Yes
Inertial navigation including horizontal velocity and distance, based on gyros and accelerometers	Yes	Yes	Yes	Yes

- Communication Bluetooth
- Application mobile « Featherweight UI » pour lecture de données et armement
- Blue Raven
- Blue Jay
- Blue Jay +
- Featherweight GPS Tracker



3 outputs programmables



Autres fonctions des outputs:

- Apogee drogue deploy
- Main chute deploy
- Apogee backup
- Main Backup
- Airstarts
- Stage separation



Featherweight Blue Raven

Featherweight Altimeters

- Avantage principal: Permet l'allumage d'un deuxième moteur
- Armement à distance
- Continuité dans les igniter
- Données temps réel (bluetooth)
- Test de vol software



Armement Magnétique



Version « non + » ne permet pas de configurer les outputs avant l'apogée

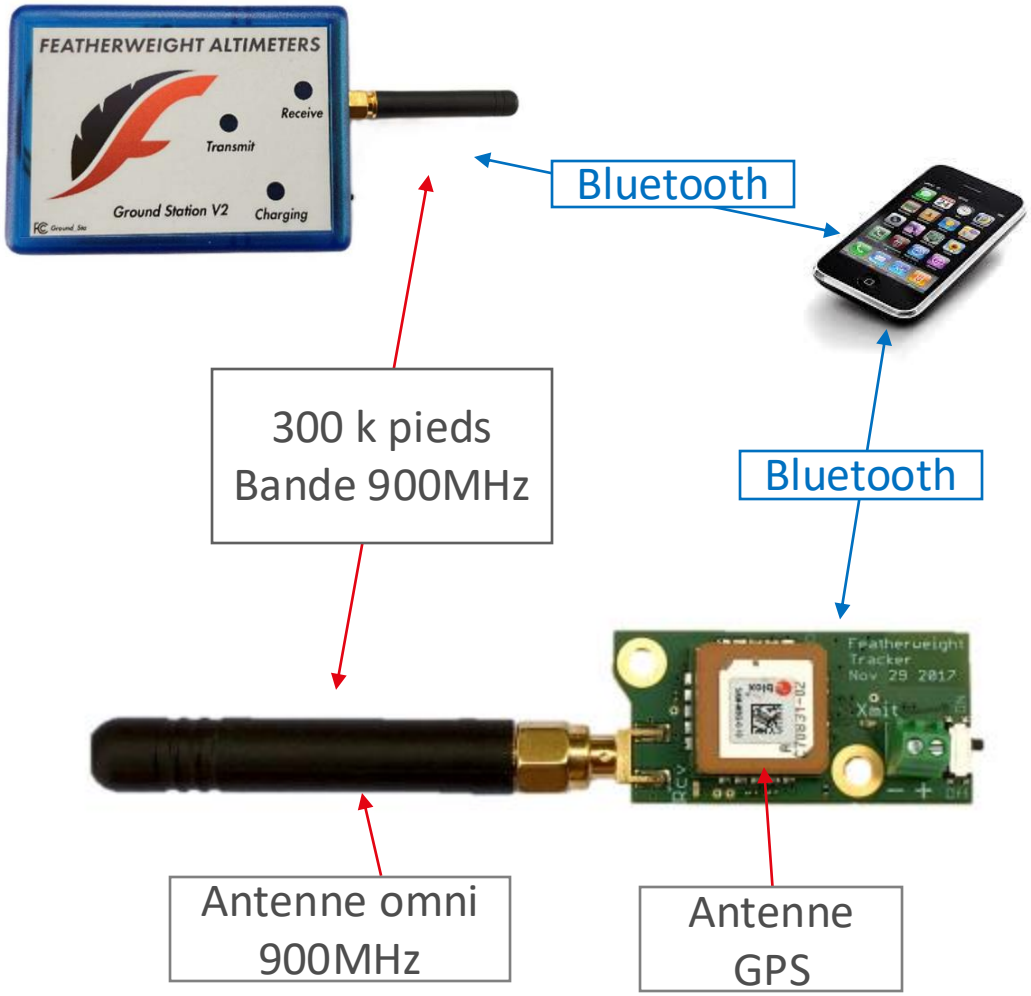
Nous possédons un exemplaire de la **version 1**

Le seul exemplaire qu'on avais est brisé

Featherweight Blue Jay+

Featherweight Altimeters

- Seulement 2 outputs configurables
- Armement à distance
- Continuité dans les igniter
- Données temps réel (bluetooth)
- Test de vol software



Featherweight GPS

Featherweight Altimeters

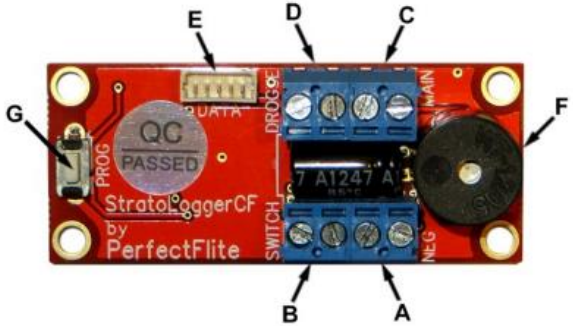
- Sélection de fréquences
- Constellations GPS, GLONASS, Galileo et BeiDou
- Fonctionnalité « Boussole »

Statut temps réel:

- FIX GPS
- Communication
- Tension batterie
- Conflits de fréquence

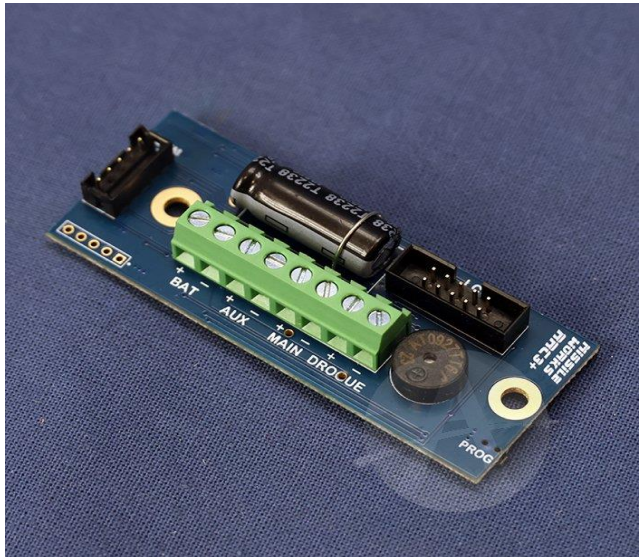
StratoLogger CF

Perfect Flite



- Ol' reliable (Utilisé dans tous nos vols depuis 2023)
- Déclenchement à l'apogée et à 1500 pieds
- Statut communiqué par séquences de buzzer
- Très difficile à acheter





Missileworks RRC3+

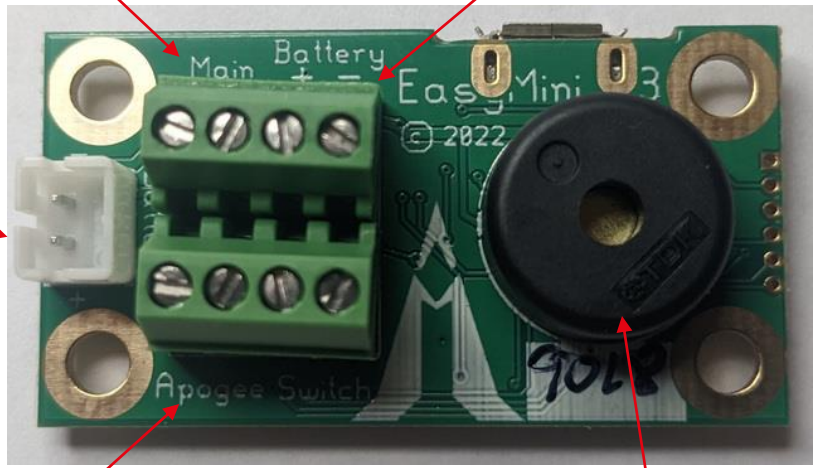
Perfect Flite

- Tentative de remplacer le Stratologger
- 2 output classique + 1 programmable
- Problème avec le logging, mais a bien fonctionné tout de même à LC 2026

Charges pour séparation à une altitude programmable (1500 pieds)

Bornier pour batterie 3.7 à 12V

Connecteur JST pour batterie Li-ion 1s (3.4-4.2V)



Charges pour séparation à l'apogée

Buzzer (pas très bruyant)

Easymini

Altus Metrum

Port micro USB pour data

LC-2023 Succès

SAC-2024 Succès

LC-2024 Succès

IREC-2025 Succès

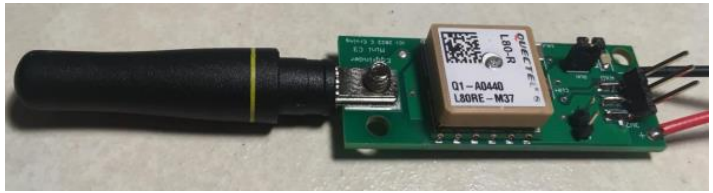
LC-2025 Erreur!! Reboot à l'apogée

Magané un peu

Eggfinder Mini Transmitter

EggTimer Rocketry

Documentation Transmitter



Peu couteux

Doit être assemblé par le client

902-928 MHz ISM band

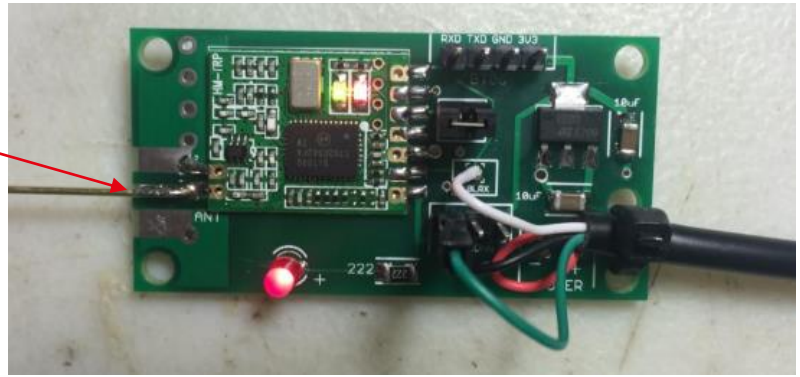
Placé dans la fusée et communique avec le dongle (prochaine page)

Eggfinder RX « Dongle » Receiver

EggTimer Rocketry
Documentation RX Dongle

- Peu couteux
- Doit être assemblé par le client (nous)
- 902-928 MHz ISM band
- Le receiver est utilisé au sol et reçoit les données du transmitter.
- Communication filaire UART vers SAS ou PC

Notre version possède un connecteur SMA et une antenne omni directionnelle



Spotrace





- Ne jamais lancer sans ça
- Trackeur de camping
- Communique par satellite à une application
- Nécessite internet pour consulter position
- Nécessite aussi abonnement mensuel

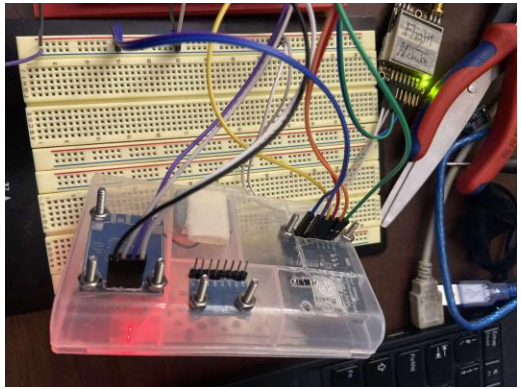




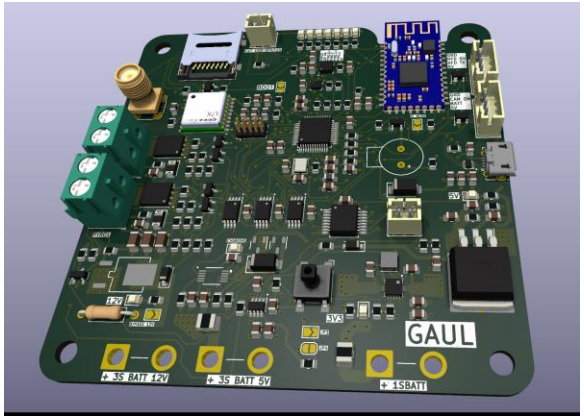
SRAD

(STUDENT RESEARCHED AND DEVELOPPED)

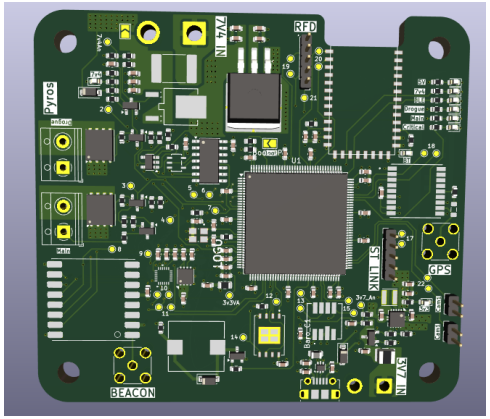
Arduino 2022-2023



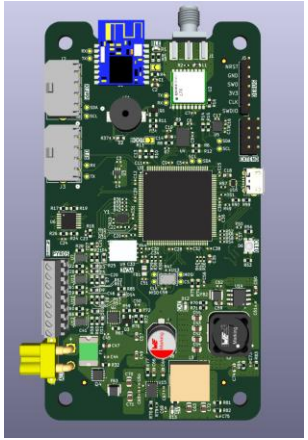
ODB1 2023-2024



ODB2 2024-2025



ODBlix 2025-2026



Ordinateur de bord

2022 à aujourd'hui…

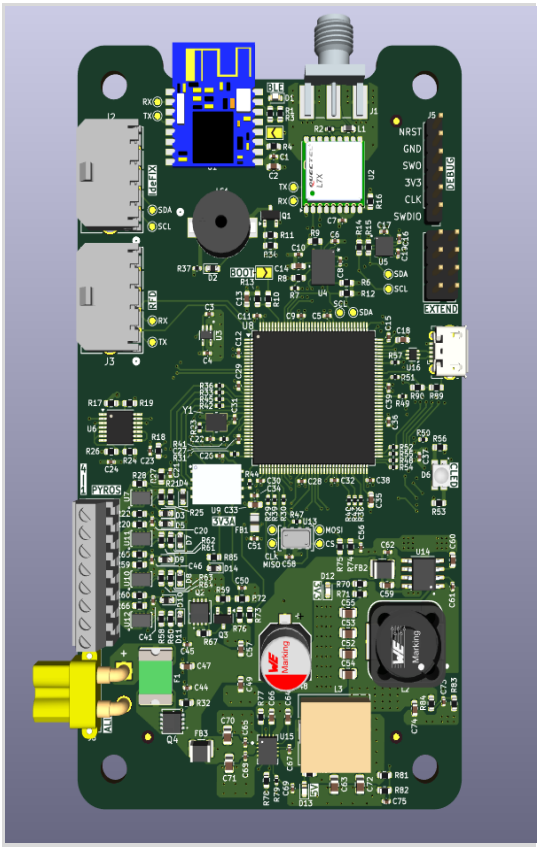
Objectif: Englober tous les requis avioniques sur un PCB robuste et équipé de capteurs précis.

Première itération:

GPS, IMU 6-axes, Baromètre, SD, Radio, Bluetooth, Pyros(2x), Buzzer, Support Caméras(2x)

Deuxième itération:

GPS, IMU 9-axes, Baromètre, Mémoire NOR, Radio, Bluetooth, Buzzer, Pyros(2x), IdeFIX, Support Caméras(2x)



Ordinateur de bord

2022 à aujourd'hui…

Objectifs 2027:

- Compatibilité > 100 000ft
- Tests plusieurs reprises
- Produit fini et programmable par Bluetooth
- Pas de V3 hardware prévue, uniquement software

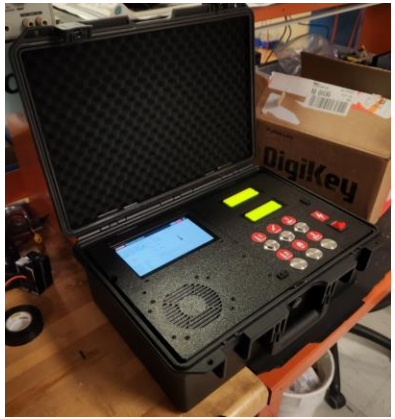
Laptop 2022-2023



SASv1 2023-2024



SASv2 2024-2025



SASv3 2025-2026

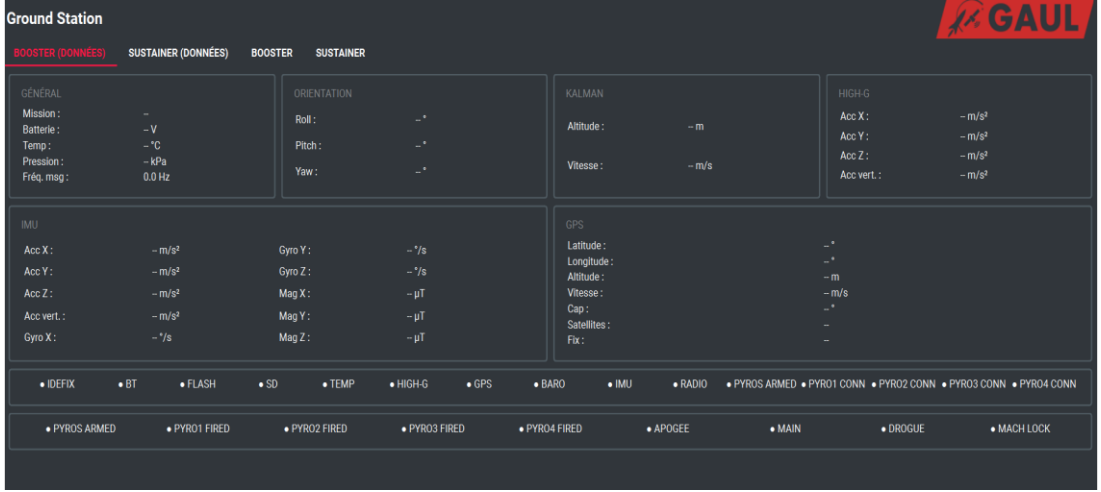


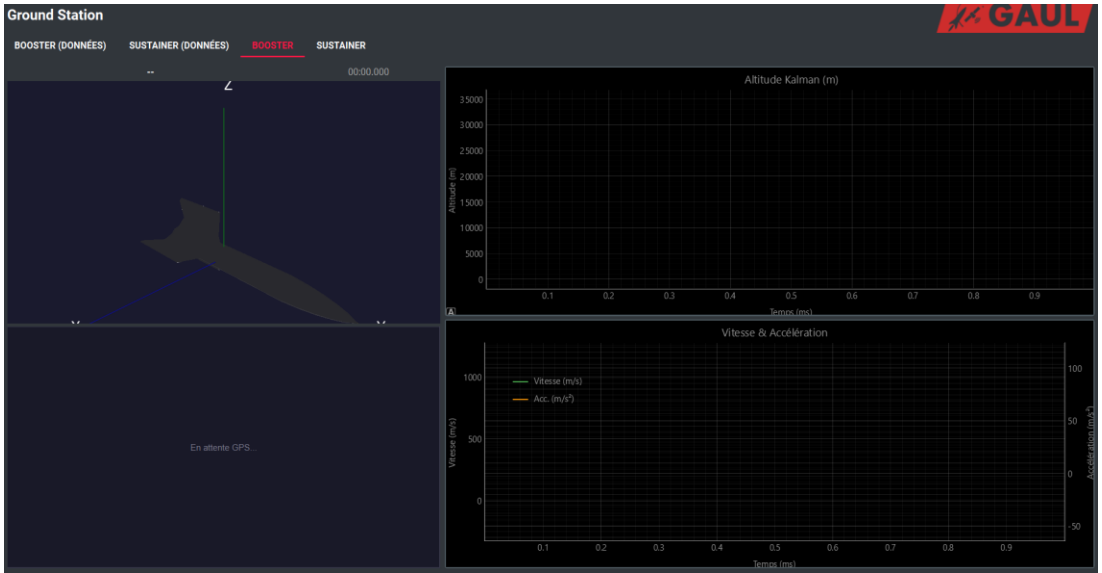
Station au sol

2023 à aujourd'hui…

Objectif: afficher les données de vol clairement pour faciliter le diagnostique de la fusée et sa récupération après le vol

Soit transitionnner à un produit plus facile à utiliser, soit revenir au laptop…



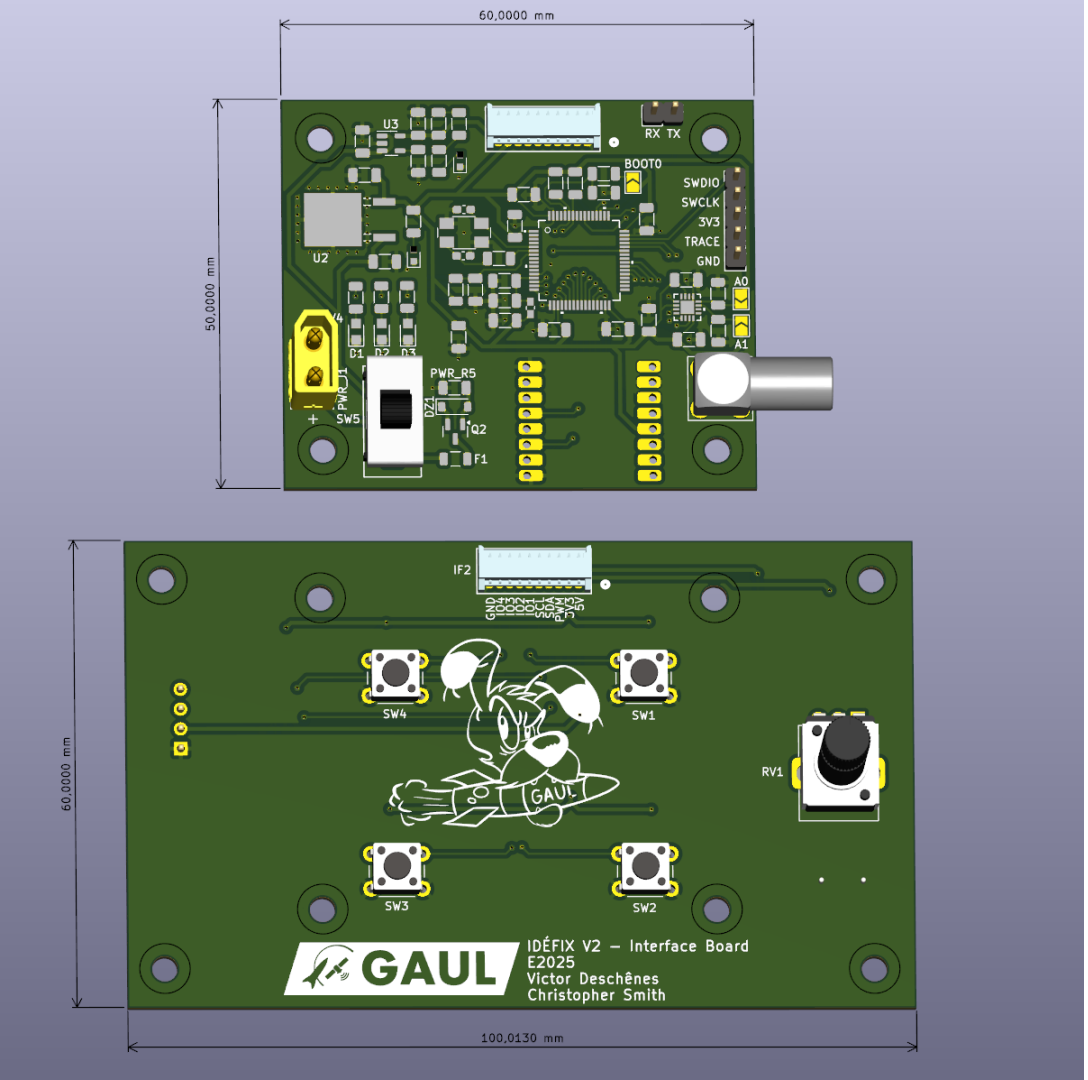


Station au sol

2023 à aujourd'hui...

Objectifs 2027:

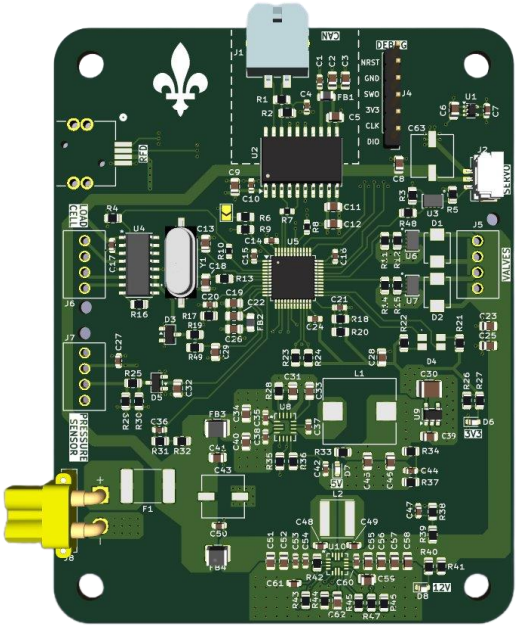
- Interface plus robuste et facile à debugger (code d'erreurs, etc)
- Revoir communication Mavlink. Possiblement implémenter à la main un protocole.

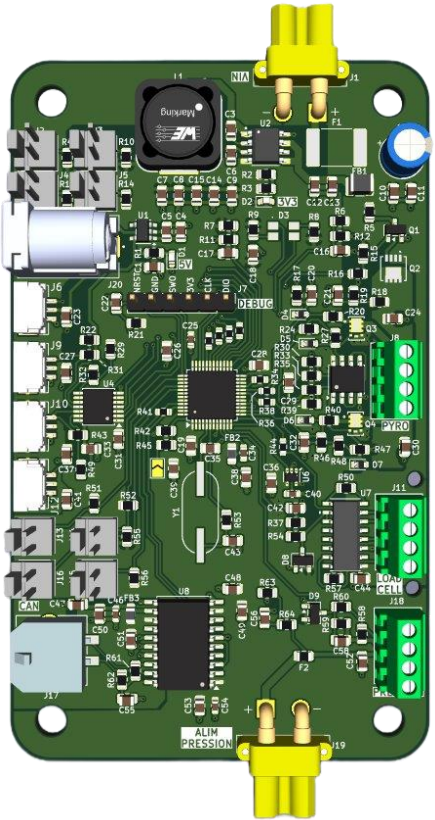


IdéFIX

Beacon et Boussole

- Opère dans la bande 433MHz
- Utilise une antenne Yagi unidirectionnelle
- Permet de « pointer » la direction de la fusée à travers des obstacles (arbres)
- Ajouter chip GPS
- Revoir fréquences



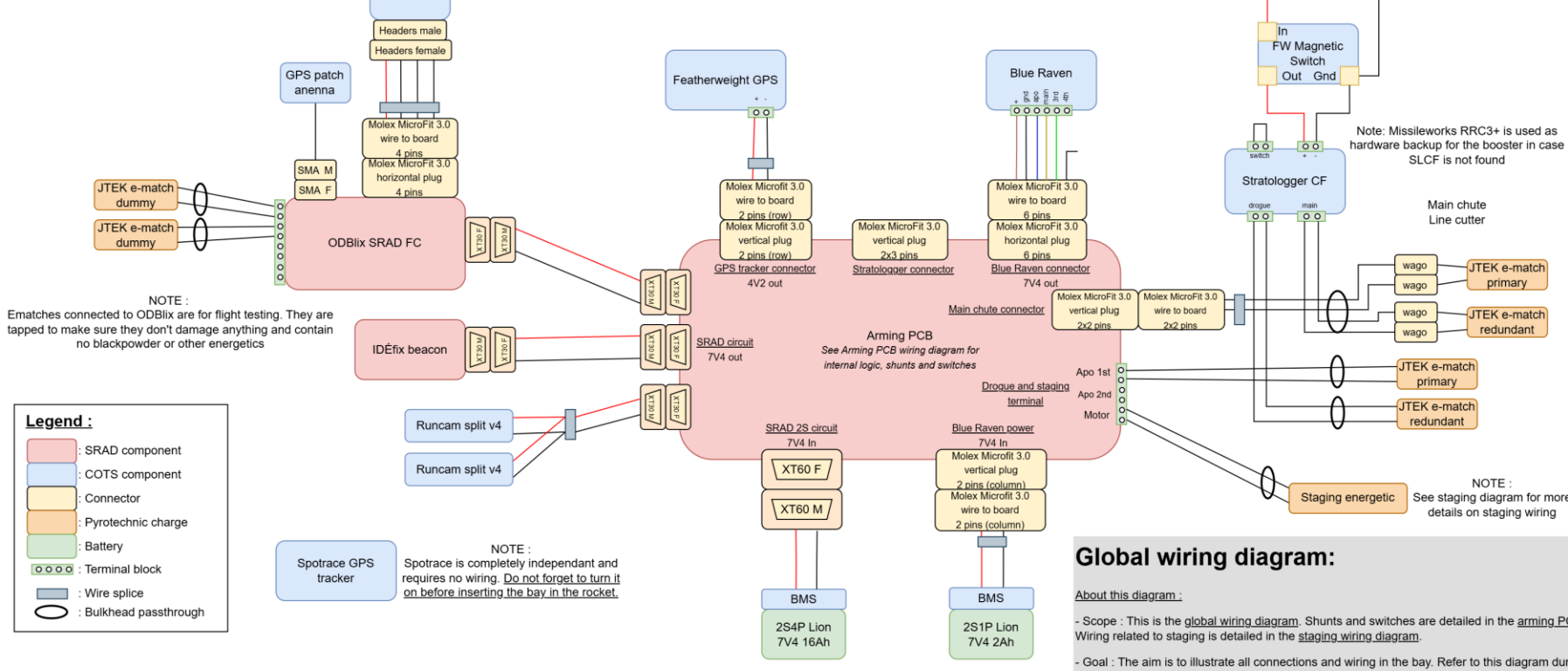


Cetautomatix

PCB moteur et GSE

- Deux PCB extrêmement rushés la session dernière
- Assemblés, non testé
- PCB moteur à refaire
- PCB GSE n'aurait même pas pu être utilisé, projet GSE industriel cette année

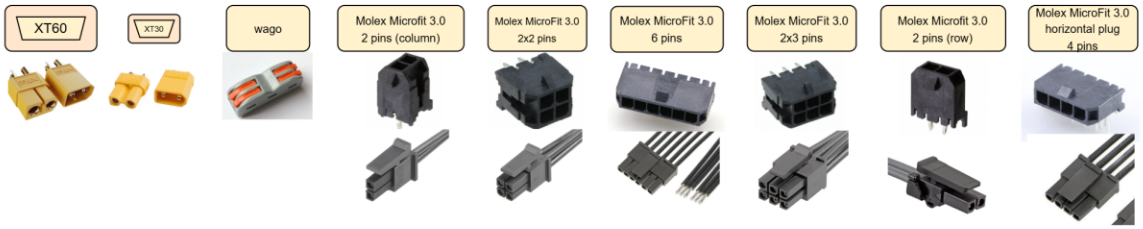
SYSTÈME LC 2026



Legend :

- : SRAD component
- : COTS component
- : Connector
- : Pyrotechnic charge
- : Battery
- : Terminal block
- : Wire splice
- : Bulkhead passthrough

Connector Identification:



Global wiring diagram:

About this diagram :

- Scope : This is the global wiring diagram. Shunts and switches are detailed in the arming PCB wiring diagram. Wiring related to staging is detailed in the staging wiring diagram.

- Goal : The aim is to illustrate all connections and wiring in the bay. Refer to this diagram during the bay's assembly.

- Stage : This diagram is valid for both the booster and sustainer's avionics bays

About shunts and switches :

- See Arming PCB diagram to see exact wiring and logic for shunts and switches

- All 5 Pyrotechnic charges are disconnected and shunted by the arming PCB

About Wire Joints :

- Connectors: they are crimped or soldered depending on the manufacturer's recommendations
    - We buy all Molex and JST wire-to-board connectors pre-crimped. We splice the wires as needed
    - XT30 and XT60 are soldered
- Splices: wires are twisted, solder is applied to solidify, and the splice is covered with shrink tube

Staging wiring diagram:

About this diagram :

- Scope : This is the staging wiring diagram. Avionics not related to staging are either not depicted or simplified. Refer to the global wiring diagram for details on the complete avionics bay.

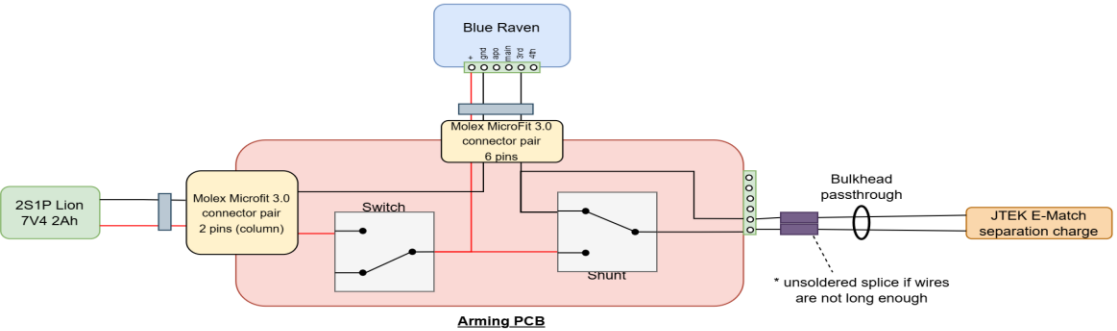
- Goal : The aim is to illustrate how staging charges are switched and deployed.

- Stage : This diagram depicts both stages

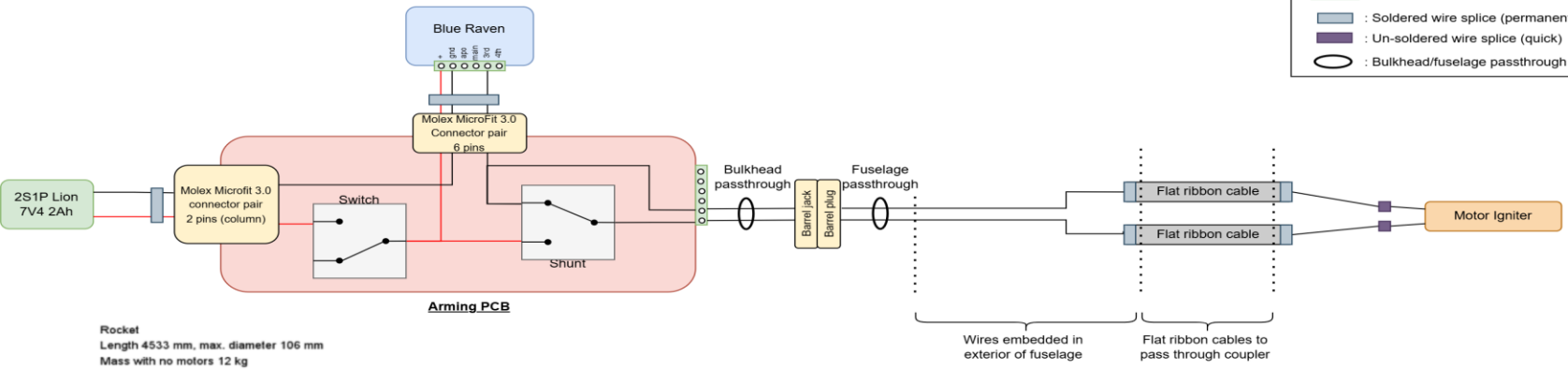
About Wire Joints :

- Connectors: they are crimped or soldered depending on the manufacturer's recommendations
    - We buy all Molex wire-to-board connectors pre-crimped. We splice the wires as needed
    - XT30 and XT60 are soldered
- Permanent splices: wires are twisted, solder is applied to solidify, and the splice is covered with shrink tube
- Quick splices: wires are twisted and covered with electric tape (only for splices made at launch site)

Booster staging



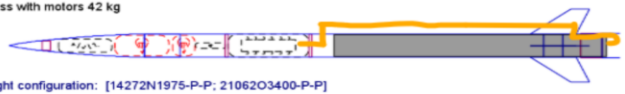
Sustainer staging



Legend :

- : SRAD component
- : COTS component
- : Connector
- : Pyrotechnic charge
- : Battery
- : Terminal block
- : Soldered wire splice (permanent)
- : Un-soldered wire splice (quick)
- : Bulkhead/fuselage passthrough

Rocket
Length 4533 mm, max. diameter 106 mm
Mass with no motors 12 kg
Mass with motors 42 kg

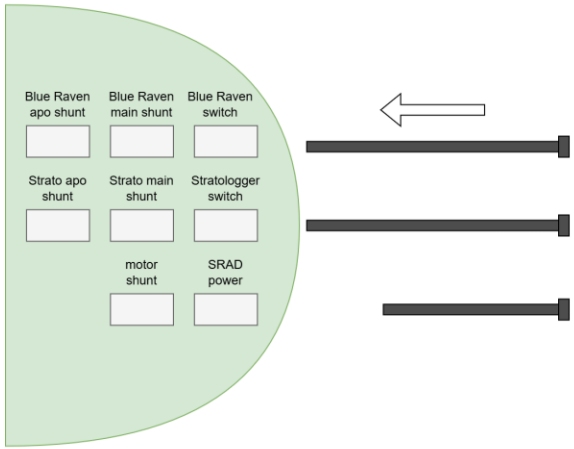


Flight configuration: [14272N1975-P-P; 21062O3400-P-P]
Apogee: 92659 ft
Max. velocity: 946 m/s (Mach 3.087)
Max. acceleration: 11.7 G

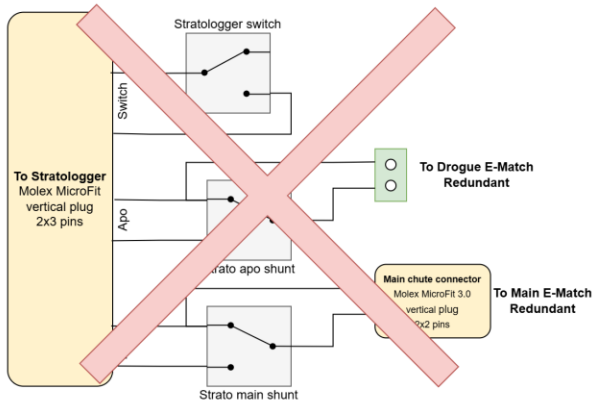
Wiring path :
- Pass through bulkhead into top of motor
- Pass through fuselage into fuselage embedded wire
- Pass through coupler via flat ribbon cables

NOTE :
A pair of flat cables is used for each igniter conductor for redundancy.
This kind of assembly was used at IREC and correctly ignited the igniter. It was also not damaged after flight.

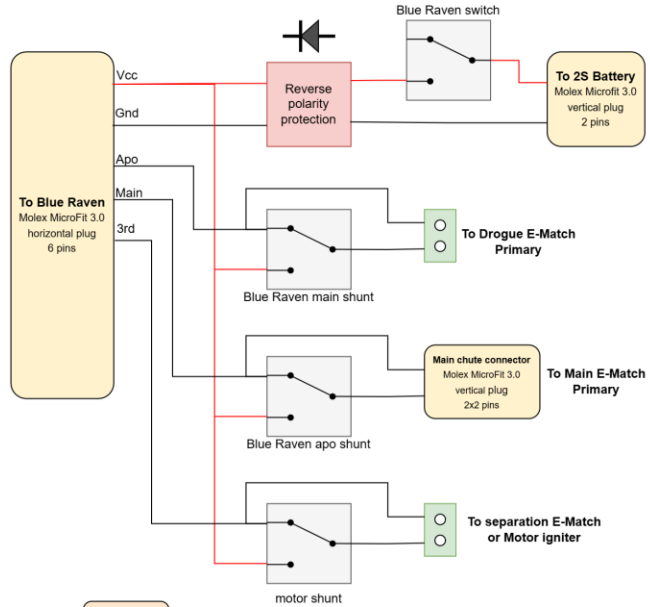
Push-pin mapping in arming PCB



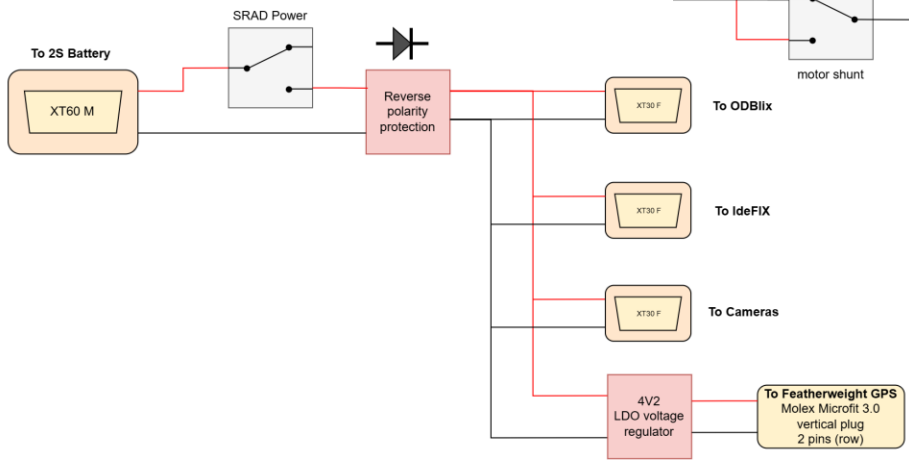
Stratologger Circuit (not used anymore)



Blue Raven Circuit



Telemetry Circuit



Arming PCB diagram:

About this diagram :

- Scope : This is the Arming PCB diagram. It shows the internal logic and wiring of the arming PCB, including shunts and switches. Refer to the global avionics diagram to see how connections are made outside of the arming PCB. Since outside connections are trivial, this diagram is enough to understand the switching and shunting logic.

- Goal : The aim is to illustrate how the arming PCB shunts and switches all the connected components. It is just a more readable schematic.

- Stage : This diagram depicts both stages

- Switch states : All switches are depicted in their normal (unpressed) state. When an arming rod is inserted and presses on a push-pin, the switch goes into its pressed state (lever connects to the other terminal).

Questions?



UNIVERSITÉ LAVAL